

DESIGN SUMMARY REPORT

1.0 INTRODUCTION

Most line-following robots rely almost exclusively on infrared sensors to detect the position of the line and react afterwards. While this approach is effective for basic navigation, it limits the robot's understanding of its own motion. The microcontroller reacts only to what the infrared sensors observe and has no knowledge of the robot's speed, angular velocity or wheel movement. These limitations become increasingly significant at higher speeds, where delayed reactions lead to overshoot, oscillation and reduced performance at corners.

2.0 DESIGN OBJECTIVES/PROCESS

2.1 DECISION MATRIX

Initially, the proposed added feature included more than just the IMU and wheel encoders, with additional ambient light sensing and battery monitoring to provide the robot with more environmental information. Because resources, such as power, PCB space and time were limited in this challenge, it was decided to reduce the complexity and power draw of the system by only implementing an IMU and wheel encoders. A decision matrix was used to assist with choosing the best solution, with metrics rated on a scale from 1 to 5.

DECISION MATRIX					
Added Feature	Performance Gain	Complexity	Power Consumption	PCB Space	Total
IMU	5	2	3	2	12
Wheel Encoders	4	3	4	2	13
Ambient Light Sensor	2	3	3	3	11
Battery Monitoring	1	2	2	3	8

2.2 ENGINEERING TRADEOFFS

Throughout the design process, several tradeoffs were made in finalizing the design to reduce complexity and power draw and increase performance.

Decision	Reason
6-axis IMU instead of 9-axis	3-axis magnetometer which provides absolute heading was unnecessary due to short time duration. Removing it reduces power consumption and code complexity
SPI instead I ² C	SPI provides higher speed transmission and full-duplex, which allows devices to transmit and receive simultaneously. Also slightly lower power consumption than I ² C
Optical wheel encoders	Optical wheel encoders were chosen over other wheel encoders due to them being compact and compatible with TT motors, with adjustable mounting and sufficient resolution for odometry.
Removed ambient light sensor	Added additional complexity and power consumption outweighed the expected navigation improvement
Removed battery monitoring	Decreased design complexity and power draw while increasing PCB space. Allowed time to focus on other features with greater impact

3.0 HARDWARE AND PCB DESIGN

The hardware and circuit design was chosen to reduce complexity and power consumption while increasing performance. A 6-axis IMU and dual optical wheel encoders were added to the robot and interfaced with the microcontroller to increase the information used for navigation.

The PCB was designed using a four-layer stack-up consisting of a top signal layer, an internal ground plane, an internal power plane and a bottom signal layer. Ground pours were also done on the top and bottom signal layers to minimize electromagnetic interference. This arrangement provides low impedance return paths for the high-speed digital signals while minimizing noise throughout the board.

3.1 IMU DESIGN

To provide the robot with awareness of its own motion, an IMU was incorporated into the added feature, which measures linear acceleration and angular velocity. The LSM6DSLTR IMU was chosen due to its low power draw, compact size, and resistance to noise. The IMU is supplied with 3.3V with bypass capacitors added on the inputs to reduce noise to the IMU from the rest of the components. SPI communication was used due to faster speed and simultaneous communication, with the various SPI communication lines connected to the microcontroller. Unused pins were either tied to ground or set as no-connects depending on information in the datasheet.

The IMU was intentionally placed near the center of the robot and away from motors to minimize errors caused by rotational acceleration due to corners and reduce electromagnetic interference. A continuous ground plane was placed directly beneath the IMU to provide a stable reference plane. The SPI communication lines were routed short to reduce signal delay, crosstalk and electromagnetic interference and the bypass capacitors were placed close to the power pins of the IMU to reduce supply voltage ripple.

3.2 WHEEL ENCODER DESIGN

To provide the robot with additional information regarding it’s speed and odometry, which cannot be directly measured by the IMU, dual OPB360L11 optical wheel encoders were added on the TT motors. These convert the mechanical rotation of the wheel into an electric signal using an IR sensor and a patterned rotating disk, providing accurate wheel speed information that complements the IMU and improves overall motion estimation.

The wheel encoder circuitry was positioned close to the mounting locations of the TT motors, minimizing trace length between the optical sensors and the PCB. The encoder outputs were routed through Schmitt-trigger inverters before reaching the microcontroller to eliminate noise and ensure clean digital transitions.

4.0 SOFTWARE ARCHITECTURE

The added features supplement the existing line-following algorithm, with the infrared sensor array still responsible for detecting the position of the line. During operation, the microcontroller continuously acquires measurements from the IMU, wheel encoders and infrared sensors. The IMU provides linear acceleration and angular velocity measurements, while the wheel encoders measure wheel speed and estimate the distance traveled through odometry. These measurements are combined with the infrared sensor data to estimate the robot’s current motion state

With this, the software estimates the severity of the upcoming corners by analyzing the line position across all seven infrared sensors while simultaneously monitoring the robot’s speed. This allows the controller to predict how aggressively the robot should slow before entering a corner rather than reacting after the turn has already begun. An adaptive PID controller then adjusts according to the estimated corner severity and robot speed. During straight sections, higher speeds can be used and as corner severity increases, the controller gradually reduces the target speed and modifies the PID gain to maintain stable tracking while minimizing overshoot.

5.0 POWER

Minimizing power consumption was a primary design objective throughout development. Several proposed features were ultimately removed in part due to their power consumption. The selected IMU was chosen due to its high performance at low power draw, while the largest power consumption comes from the infrared emitters within the wheel encoders, which are required for reliable optical speed measurement. Overall, the completed added feature consumes approximately 163 mW of power, representing only a small portion of total power consumption. The breakdown can be seen in the table below.

POWER CALCULATIONS				
Component	Current (Typical)	Power	Quantity	Total
LSM6DSLTR (IMU)	0.65 mA	2.145 mW	1	2.145 mW
OPB360L11 (LED)	14 mA	46.2 mW	2	92.4 mW
OPB360L11 (Phototransistor)	10 mA	33 mW	2	66 mW
150060VS75000 (Green LED)	0.1 mA	0.33 mW	2	0.66 mW
SN74HC14DR (Schmitt Trigger Inverter)	0.4 mA	1.32 mW	1	1.32 mW
Total Added Power				162.53 mW

6.0 EXPECTED PERFORMANCE IMPROVEMENTS

The added feature is expected to improve both the speed and accuracy of the line-following robot by providing additional information regarding the robot’s motion. Rather than relying solely on infrared sensors, the controller gets information about angular velocity, linear acceleration, wheel speed and estimated travel distance. This addition information enables predictive corner detection, adaptive speed planning and dynamic PID tuning. As a result, the robot is expected to maintain higher speeds along straight sections while reducing speed before entering sharp corners. This is expected to reduce overshoot, improve tracking consistency and decrease time.